Supplemental Instruction Lesson Plan

Session Date: 3 MAR 2022

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Course: CSCI160 (Computer Science I)

Instructor: Ken Whitener

Lesson: Data Validation

Background: Up to this point, we have been writing code that takes user input, and hoping that the users will provide the input in a format that we expect. However, computer users will usually just provide input however they want and hope that the program is smart enough to accept it or correct it. If programmers did not validate user input, then software is liable to crash, which can be very frustrating for a user. This week we have learned about Boolean Logic, If-Statements, and the Conditionals that go with If-Statements to help computers make decisions. With these tools, we are well equipped to check user-input for correct formatting.

Practical Applications/Skills:

- Comparison Operators
- Relational Operators
- Decisions and Boolean Conditions
- UI/UX (User Experience)

Agenda:

1. Demo
   I. Refresh String Split
   II. Refresh Indexing
   III. Conditions using Data Structures
2. Group Activity

1. Demo

I. Refresh String Split

II. Refresh Indexing

III. Conditions using Data Structures

2. Group Activity

I. Date Validation

a. Form groups of two. In paired programming, there are typically two roles: Driver and Navigator. The Driver will be writing the code, while the navigator focuses on checking the accuracy of the code written by the Driver. Come to an agreement with your partner on how to distribute the roles (Decide what works best for the team to maximize team growth. If you feel uncomfortable being the driver, then try being the driver.)

b. Ken has provided a template to start the code with instructions. Open and edit the attached "date_validation_handout.py" file. Complete the program to accept a user input in a date format. Provide print statements to let the user know if their input was correct. Be creative with the outputs, as long as it demonstrates and conveys that it knows if the format was correct. The format/rules to apply are listed below:

Program will ask for a date in MM/DD/YYYY and determine if the format and values are correct.

The following should be tested

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~ Are there three fields?

~ Are month and day a 2-digit integer?

~ Is the year a 4-digit integer

~ Is the month within the correct range {1 - 12}

~ Is the number of days correct for the month

*We will NOT handle leap years yet so just use 02/28/XXXX for the maximum allowed value for Feb

Example Input: 1/1/2022 Example Output: This is a valid date in the correct format	Example Input: 2/31/2022 Example Output: This is not a valid date or not in the correct format 31 is not a valid day in February	Example Input: 6/10 Example Output: This is not a valid date or not in the correct format You did not provide 3 fields separated by slashes